



NATIONAL UNIVERSITY OF MODERN LANGUAGES

VISUAL PROGRAMMING Final Lab Exam

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PROGRAM: BSSE (6th Semester)

COURSE: Visual Programming

SUBMITTED TO: Sir M Irtaza

DATE: 5/4/2026

DAY: Tuesday

Form Layouts

Form1

Name

Date of birth

Date of joinig

Form2

	Id	Name	DateOfBirth	DateOfJoining	YearsOfService	Salary
▶	1	Rafay Khan	6/22/2000 6:47 ...	6/25/2019 6:47 ...	6	500000.00
	2	Abdul Rafay	6/6/2000 6:48 PM	5/4/2026 6:48 PM	0	250000.00
	3	I am Rafay	10/4/2004 6:49 ...	5/23/2023 6:49 ...	2	250000.00
	4	Rafay khan	10/4/2004 6:49 ...	10/3/2019 6:49 ...	6	500000.00

Sql Workbench Data

SE-DEVELOPLAB33\S...mployeeSeniorities		SE-DEVELOPLAB33\...y - dbo.Employees		
	Id	EmployeeId	YearsOfService	Salary
▶	1	1	6	500000.00
	2	2	0	250000.00
	3	3	2	250000.00
	4	4	6	500000.00
*	NULL	NULL	NULL	NULL

SE-DEVELOPLAB33\...y - dbo.Employees		SE-DEVELOPLAB33\S...mployeeSeniorities		SE-DEVELOPLAB33\...2 - dbo.Employees
	Id	Name	DateOfBirth	DateOfJoining
▶	1	Rafay Khan	2000-06-22 18:4...	2019-06-25 18:4...
	2	Abdul Rafay	2000-06-06 18:4...	2026-05-04 18:4...
	3	I am Rafay	2004-10-04 18:4...	2023-05-23 18:4...
	4	Rafay khan	2004-10-04 18:4...	2019-10-03 18:4...
*	NULL	NULL	NULL	NULL

Form1.cs

```
using System;

using System.Collections.Generic;

using System.ComponentModel;

using System.Data;

using System.Drawing;

using System.Linq;

using System.Text;

using System.Threading.Tasks;

using System.Windows.Forms;

namespace WindowsFormsApp3
{
    public partial class Form1 : Form
    {
        public Form1()
        {
            InitializeComponent();
        }

        private void Form1_Load(object sender, EventArgs e)
```

```
{

}

private void label1_Click(object sender, EventArgs e)
{

}

private void button1_Click(object sender, EventArgs e)
{
    using (var db = new AppDbContext())
    {
        db.Database.CreateIfNotExists();
        var emp = new Employee
        {
            Name = textBox1.Text,
            DateOfBirth = dateTimePicker1.Value,
            DateOfJoining = dateTimePicker2.Value
        };

        db.Employees.Add(emp);
        db.SaveChanges();

        var seniority = EmployeeService.CalculateSeniority(emp);

        db.EmployeeSeniorities.Add(seniority);
        db.SaveChanges();

        MessageBox.Show("Employee Saved Successfully!");

        Form2 form2 = new Form2();
        form2.Show();
    }
}
```

```
}  
}
```

Form 2.cs

```
using System;  
using System.Linq;  
using System.Windows.Forms;  
  
namespace WindowsFormsApp3  
{  
    public partial class Form2 : Form  
    {  
        public Form2()  
        {  
            InitializeComponent();  
        }  
  
        private void Form2_Load(object sender, EventArgs e)  
        {  
            using (var db = new AppDbContext())  
            {  
                var data = db.Employees  
                    .Select(emp => new  
                    {  
                        emp.Id,  
                        emp.Name,  
                        emp.DateOfBirth,  
                        emp.DateOfJoining,  
  
                        Years = emp.Seniorities.Select(s => s.YearsOfService).FirstOrDefault(),  
                        Salary = emp.Seniorities.Select(s => s.Salary).FirstOrDefault()  
                    })  
                    .ToList();  
  
                dataGridView1.DataSource = data;  
            }  
        }  
    }  
}
```

```

    }

    private void dataGridView1_CellContentClick(object sender, DataGridViewCellEventArgs e)
    {

    }

}

}

```

Employee.cs

```

using System;

using System.Collections.Generic;

using System.ComponentModel.DataAnnotations;

public class Employee
{
    [Key]

    public int Id { get; set; }

    public string Name { get; set; }

    public DateTime DateOfBirth { get; set; }

    public DateTime DateOfJoining { get; set; }

    public virtual ICollection<EmployeeSeniority> Seniorities { get; set; }

}

```

Employee Seniority.cs

```

using System.ComponentModel.DataAnnotations;

using System.ComponentModel.DataAnnotations.Schema;

public class EmployeeSeniority
{
    [Key]

    public int Id { get; set; }

    [ForeignKey("Employee")]

    public int EmployeeId { get; set; }

    public int YearsOfService { get; set; }

    public decimal Salary { get; set; }

    public virtual Employee Employee { get; set; }
}

```

```
}
```

Employee Service.cs

```
using System;
```

```
public class EmployeeService
```

```
{
```

```
    public static EmployeeSeniority CalculateSeniority(Employee emp)
```

```
    {
```

```
        int years = DateTime.Now.Year - emp.DateOfJoining.Year;
```

```
        if (emp.DateOfJoining > DateTime.Now.AddYears(-years))
```

```
            years--;
```

```
        decimal salary = 0;
```

```
        if (years >= 5)
```

```
            salary = 500000;
```

```
        else
```

```
            salary = 250000;
```

```
        return new EmployeeSeniority
```

```
        {
```

```
            EmployeeId = emp.Id,
```

```
            YearsOfService = years,
```

```
            Salary = salary
```

```
        };
```

```
    }
```

```
}
```